INSTANTONS & 3-MANIFOLDS - Fall 2026 - IRVING DAI

Lecture 1 // Aug 25 // Overview

**Thm** [Freedman]

$Q$ = unimodular symmetric bilinear form

$\exists!$ $\pi_1 = 1$ Top $X^4$ s.t. $Q_X = Q$

1) $Q$ even, $\exists!$ SM $X^4$ / homeo

2) $Q$ odd, $\exists$ at most 2 SM $X^4$ / homeo

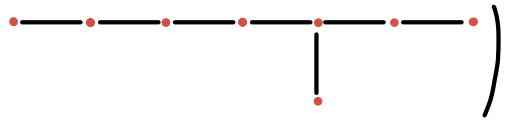
of which at most 1 is smoothable

**Thm** [Donaldson]

If $X^4$ is $\pi_1 = 1$, smooth, closed 4-mfd and $Q_X$ is definite,

then $Q_X$ is equivalent to $n(+1)$ or $n(-1)$

**Ex** $E_8$ matrix:

adj (  ) $=$

$$\begin{bmatrix} -2 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & -2 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & -2 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & -2 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & -2 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 & -2 \end{bmatrix}$$

Check: no change of basis to $\pm I_d^8$.

$\Rightarrow$ $E_8$ realized by non-smoothable top $X^4$

$\Rightarrow$ Same for $nE_8$

Over the next few weeks, we'll discuss the pf of Donaldson.

After, we'll discuss IFH and some applications to three manifolds, in particular, the homology cobordism group. The rest of this lecture will be to lay the groundwork for this.

## 3-Manifolds

**Def** Homology Cobordism Group

$\Theta^3_{\mathbb{Z}} = \{\text{all } \mathbb{Z}HS^3\text{'s}\}$ / homology cobordism

$Y_1 \sim Y_2$ hx-cob if $\exists\ W^4$ a homology $S^3 \times I$ st $\partial W^4 \cong -Y_1 \sqcup Y_2$

↳ TOP version is trivial

**Thm** [Freedman]

If $Y$ is a $\mathbb{Z}HS^3$, then $Y = \partial W$ for <u>some</u> $\mathbb{Z}HB^4$

In fact, $\pi_1 W = 1$

**Pf** Given unim. sym. bil. pairing $Q$, can construct sm $X^4$ with $\partial$ w/ intersection form $Q$ by attaching 2-h's to $B^4$ according to $Q$

$E_8$:



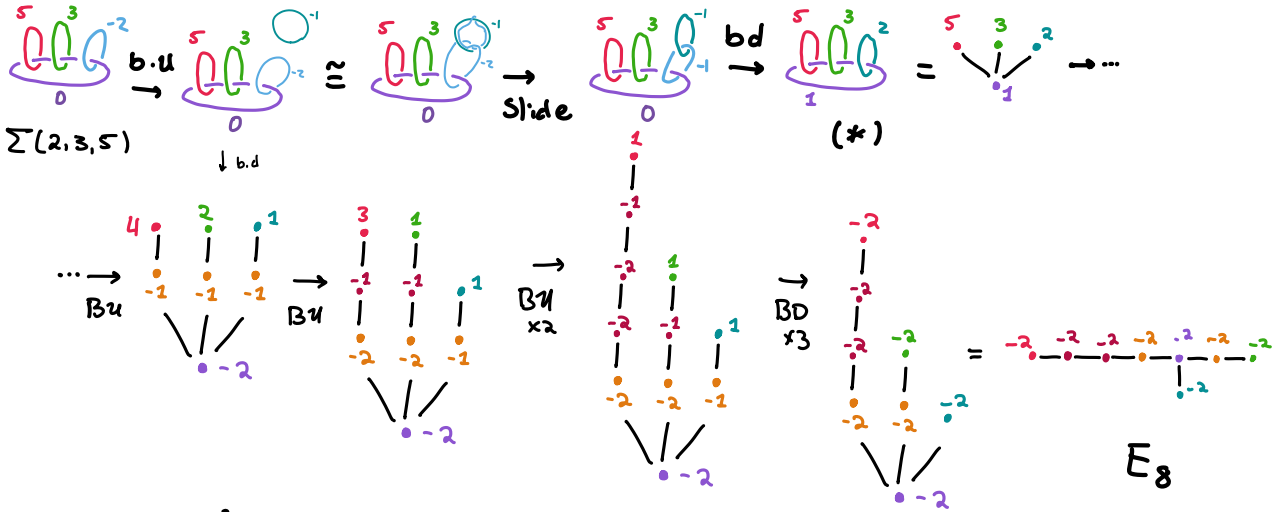
For $E_8$, $\partial E_8 \cong$ Poincaré hx sphere $\cong \Sigma(2,3,5)$

By Donaldson, $\Sigma(2,3,5)$ does not bound any smooth $\mathbb{Z}HB^4$

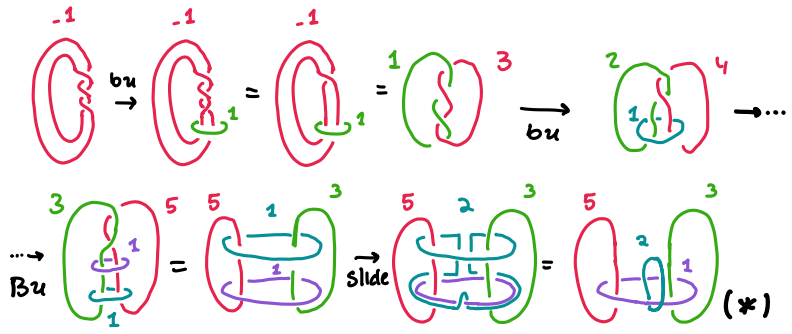
$\Rightarrow [\Sigma(2,3,5)] \neq 1 \in \Theta^3_{\mathbb{Z}}$

Pf of $\partial$-diffeo:

$\Sigma(2,3,5) \cong \partial E_8$




Poincaré Hx $S^3 \cong \Sigma(2,3,5)$



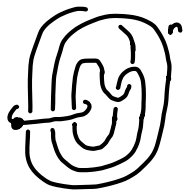
Rmk There are many $\mathbb{Z}HS^3$'s

Ex Brieskorn Spheres: $p, q, r$ pairwise coprime

$\Sigma(p,q,r) := S^5_\varepsilon \cap \{x^p + y^q + z^r = 0\} \subseteq \mathbb{C}^6$

Ex $S^3_{1/n}(k)$ $\forall\, k$

Ex Mazur Mfds: $1\text{-}h^4 \cup 2\text{-}h^4$ s.t. alg linking is $\pm 1$, geom linking $\neq \pm 1$



Q How large is $\Theta^3_{\mathbb{Z}}$?

Thm [Furuta, FS]

$\{\Sigma(2,3,6n-1)\}$ is a $\mathbb{Z}^\infty$ subgp of $\Theta^3_{\mathbb{Z}}$.

Thm [Daiemi-Nozaki-Sato-Taniguchi]

sometimes $\{S^3_{1/n}(k)\}$ are lin indep.

↑ Specific condition on $\mathit{IFH}_*$

Note $\Sigma(2,3,6n-1) = S^3_{1/n}(T_{2,3})$

From concordance to Hx cobordism

$\Sigma_2 : \mathcal{C} \longrightarrow \Theta^3_{\mathbb{Z}/2}$

$K \longmapsto \Sigma_2(K)$ = Double branched cover over $K$

is a homomorphic

Q Is there a distinction between bounding a $\mathbb{Z}HB^4$ and a $\pi_1 = 0$ $\mathbb{Z}HB^4$?

Ie, is there a $\pi_1 = 1$ hx cob from $\Sigma(2,3,5)$ to itself?

Thm [Taubes]

No.

Thm [Daiemi-Nozaki-Sato-Taniguchi]

often no

cut open cobordism ⇝ get s.c. $\mathbb{Z}HB^4$ bounding $Y^3 \# -Y^3$